INTERNSHIP REPORT ON
HOUSE PREDICTION SYSTEM

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ಆಗು ನೀ ಅನಿಕೇತನ
BE BOUNDLESS
DR. MANMOHAN SINGH BENGALURU CITY UNIVERSITY

Submitted In Partial Fulfilment of The Requirements of The Bachelor of Computer Application Degree Course of Dr. Manmohan Singh Bengaluru City University.

BY

SYED ABDUL KHADER NIZAMI

REG NO: U18EP23S0034

BATCH 2023-2026

Under the Guidance of

Prof. MANOJ.P

Department of Computer Application MEWA VANGUARD

BUSINESS SCHOOL

MEWA VANGUARD BUSINESS SCHOOL
(Affiliated to Dr. Manmohan Singh Bengaluru City University & Approved by AICTE)

Mewa Vanguard Business School

Jayanagar 9th Block, Bengaluru

Karnataka - 560069

Affiliated to Dr. Manmohan Singh Bengaluru City University & Approved by AICTE

Evaluation OF INTERNSHIP REPORT

For submission to Dr. Manmohan Singh Bengaluru City University.

Student	Mentor/Guide
NAME: SYED ABDUL	**Name: manoj.**
Reg no: U18EP23S0034	Signature:
Signature:	Date:
Date:	

EVALUATION SUMMARY

	PROJECT	VICE-VOCE	TOTAL
MAX. MARKS	60	40	100
MARKS ALLOTTED			
NAME OF EXAMINER:		Signature:	

COLLEGE CERTIFICATE

This is to certify that the internship report on "HOUSE PRICE PREDICTION SYSTEM" was carried out by ***SYED ABDUL KHADER NIZAMI*** *a Student of the VI Semester BCA, bearing Reg No. U18EP23S0034 has completed his 8-week organizational study.*

The Internship report is submitted to the Dr. Manmohan Singh Bengaluru City University in the Academic Year 2023-26, in partial fulfillment for successful completion of the VI semester BCA and award of a Bachelor of Computer Applications Degree from Dr. Manmohan Singh Bengaluru City University.

PROJECT GUIDE	**PRINCIPAL**
Prof. manoj.p	BHARTHISH RAO R. S

Place: Bengaluru Date:

DECLARATION BY THE STUDENT

I hereby declare that this Internship report "HOUSE PRICE PREDICTION SYSTEM" is an original record of my experience while conducting the internship study by 'EDU TECH'

The organizational study was carried out by me, in partial fulfillment for the successful completion of the VI semester BCA of Dr. Manmohan Singh Bengaluru City University. This report is not submitted to any other university for the award of any Diploma/Degree etc.

Register No.	Student Name	Signature
U18EP23S0034	***SYEDABDUL KHADER NIZAMI***	

Place: Bengaluru Date:

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LOG SHEET

Week	Date	Activities Performed	Mentor Signature
Week 1		Orientation, Python basics, NumPy & Pandas introduction	
Week 2		Data collection techniques, exploratory data analysis (EDA)	
Week 3		Data preprocessing: handling missing values, normalization	
Week 4		Feature engineering, Linear Regression theory & implementation	
Week 5		Model training & hyperparameter tuning, model evaluation	
Week 6		Flask web app development, integrating ML model with UI	
Week 7		Testing, debugging, and UI improvement	
Week 8		Project documentation, final presentation, submission	

INTERNSHIP COMPLETION CERTIFICATE

This is to certify that SYED ADUL, bearing Registration No. U18EP23S0047, has successfully completed an 8-week internship program at EDU TECH India Pvt. Ltd. in the domain of Data Science and Machine Learning.

During the internship period, the student demonstrated sincere dedication, technical aptitude, and a commitment to learning. The student worked on real-world data science projects, including the development of a House Price Prediction System using Linear Regression.

We wish the student all the best in future endeavors.

Authorized Signatory
EDU TECH India Pvt. Ltd.
Date:

ACKNOWLEDGEMENT

I am deeply indebted to "EDU TECH" for giving me an opportunity to carry out the organizational study and for giving vital support and guidance to complete this endeavor.

I would like to express my gratitude to our principal Dr. BHARATISH RAO R. S for his constant support and encouragement.

I would like to thank our Head of Department, Neeraja Lakshmi for her constant support and encouragement.

I am extremely grateful to my guide, Prof. manoj, Department of Computer Applications and ANEESH SHETTY, NICT Mentor, for their support and guidance provided for the completion and preparation of this report. These endeavors would not have been possible without the support of my parents. My special thanks to them.

Place: Bengaluru Date:

INTERNSHIP ATTENDANCE CERTIFICATE

This is to certify that SYED ADBUL (Reg No. U18EP23S0034) has attended the internship program at EDU TECH India Pvt. Ltd. regularly and has fulfilled the required attendance criteria as per institutional norms.

The student was present for the full duration of the 8-week internship program and actively participated in all training sessions, workshops, and project activities.

HR Manager / Coordinator

EDU TECH India Pvt. Ltd.

Date:

ORGANIZATIONAL PROFILE

Edu Tech India Pvt. Ltd. is a leading edtech organization focused on delivering innovative, technology-driven educational solutions across India. The company specializes in providing practical, industry-oriented learning experiences in the fields of information technology, engineering, and professional skill development. With a strong emphasis on hands-on training and real-world applications, Edu Tech India caters to students, educators, institutions, and working professionals aiming to enhance their skills and remain competitive in the rapidly evolving digital landscape.

Edu Tech India operates as a comprehensive educational solutions provider, offering a diverse portfolio of products and services across domains such as smart classrooms, robotics and STEM education, data analytics, digital learning platforms, and hardware and networking solutions. Its programs are designed and delivered by experienced professionals, subject matter experts, and industry practitioners, ensuring learners acquire practical knowledge and job-ready competencies.

As a pioneer in technology-enabled education, Edu Tech India equips learners with the critical skills required in today's innovation-driven economy. Through its blended learning approach, modern infrastructure, and accessible digital resources, the company promotes both academic excellence and professional growth.

From customized institutional solutions to advanced training modules aligned with current industry standards, Edu Tech India continues to play a significant role in transforming traditional education systems into dynamic, future-ready learning environments.

- **Nature of business:** Edu Tech solutions and training provider
- **Company Directors:** Parveen Shameem and Saad Faisal
- **Registered Address:** Chennai, Tamil Nadu, India
- **Legal Status of the Firm:** Private Limited Company

INTERN JOB ROLE DESCRIPTION

Job Role: DATA SCIENCE INTERN

Key Responsibilities:

6.1 Learning and Skill Development:

- Participated in structured training sessions to understand the fundamentals of data science and machine learning.
- Learned Python programming and libraries such as NumPy, Pandas, Matplotlib, and Scikit-learn for data analysis and visualization.
- Gained knowledge of machine learning algorithms, especially Linear Regression for regression problems.
- Understood data preprocessing techniques such as data cleaning, handling missing values, feature scaling, and feature selection.
- Became familiar with tools like Jupyter Notebook and version control systems such as Git and GitHub.

6.2 Data Collection and Preprocessing:

- Collected real estate and house pricing datasets from reliable sources for analysis.
- Cleaned and organized data by handling missing values and removing inconsistencies.
- Performed data transformation and normalization (StandardScaler) to prepare the dataset for model training.
- Selected relevant features such as area, number of bedrooms, bathrooms, floors, garage capacity, age, location score, and school rating.
- Encoded and scaled feature data into numerical format suitable for Linear Regression.
- Split the dataset into training (80%) and testing (20%) sets.

6.3 Model Development and Implementation:

- Worked on developing a House Price Prediction System using Linear Regression.
- Implemented the Linear Regression model using Scikit-learn to predict house prices based on property features.
- Trained the model on 1600 samples and tested it on 400 samples.
- Fine-tuned feature selection and preprocessing pipelines to improve model performance.
- Integrated the trained model and scaler into a Flask web application.

6.4 Model Evaluation and Testing:

- Evaluated model performance using metrics such as R-squared (R^2 = 95.35%), Mean Absolute Error (MAE), and Mean Squared Error (MSE).
- Tested the model with unseen house data to ensure reliability and generalizability.
- Analyzed feature importance by examining Linear Regression coefficients.

- Generated price confidence ranges (±8%) and value categories (Budget/Mid-Range/Premium/Luxury).

6.5 Data Handling and Integration:

- Prepared datasets by organizing and structuring property data for model training and testing.
- Performed feature engineering to enhance model effectiveness.
- Integrated the trained Linear Regression model into a Flask web application with a dark futuristic UI.
- Built RESTful API endpoints for prediction, model info retrieval, and chart data generation.

6.6 Project Collaboration and Documentation:

- Collaborated with team members to discuss model improvements and project development.
- Documented the entire project including data sources, methodology, API routes, and results.
- Prepared reports and presentations explaining the working and outcomes of the prediction system.
- Maintained version control using Git and GitHub to track changes and updates.
- Participated in regular review meetings and incorporated feedback.

6.7 Qualifications:

- Currently pursuing a degree in Computer Applications or related field.
- Basic understanding of Python programming and statistics.
- Interest in data analysis and machine learning techniques.
- Strong analytical thinking and problem-solving skills.
- Good communication and teamwork abilities.

6.8 Benefits:

- Gained hands-on experience in machine learning and real estate data analysis.
- Learned practical implementation of Linear Regression in real-world pricing problems.
- Exposure to real-time datasets and complete model deployment through a Flask web app.
- Improved technical, analytical, and documentation skills.

PROGRAMMING LANGUAGES / TECHNOLOGIES

- Python

Python is the main programming language used in this project. It is simple, easy to learn, and widely used in data science and machine learning. It is used for data processing, model building, and generating predictions. Python also supports multiple libraries which make development faster and more efficient.

- Pandas

Pandas is used for data handling and analysis. It helps in organizing the dataset into structured tables (DataFrames). It is also used for cleaning data, handling missing values, filtering records, and preparing the dataset for model training.

- NumPy

NumPy is used for numerical computations. It helps with performing mathematical calculations on large datasets quickly and efficiently. It provides array operations which are fundamental in feeding data to machine learning models.

- Matplotlib

Matplotlib is used for data visualization. It helps in creating charts and graphs such as scatter plots and bar charts that represent the relationship between property features and prices. These visualizations aid in understanding data patterns and model behavior.

- Scikit-learn

Scikit-learn is the primary machine learning library used in this project. It provides the LinearRegression class, StandardScaler for feature normalization, and metrics for evaluating model performance such as R-squared score and mean squared error.

- Flask

Flask is a lightweight Python web framework used to build the web application. It manages URL routing, handles HTTP requests, and serves the HTML templates. Flask connects the trained ML model to the browser-based user interface through RESTful API endpoints.

- HTML / CSS / JavaScript

HTML structures the web interface, CSS styles the dark futuristic UI, and JavaScript handles dynamic interactions such as form submissions, AJAX calls to the prediction API, and rendering Chart.js visualizations in the browser.

- Chart.js

Chart.js is a JavaScript library used for creating interactive charts on the frontend. In this project it renders a Price vs. Area scatter/line chart and a Feature Impact horizontal bar chart, giving users visual insight into the prediction model.

- Pickle

Pickle is used to serialize and deserialize Python objects. The trained Linear Regression model and the StandardScaler are saved as model.pkl and scaler.pkl files using Pickle, allowing them to be loaded and reused by the Flask application without retraining.

SOFTWARE AND HARDWARE USED

8.1 Software:

Software / Tool	Version	Purpose
Python	3.10+	Core programming language
Flask	3.0.0	Web framework for backend
Scikit-learn	1.4.0	Linear Regression & evaluation
Pandas	2.1.4	Data manipulation & analysis
NumPy	1.26.3	Numerical computations
Matplotlib	3.8+	Data visualization
Pickle	Built-in	Model serialization
VS Code / Jupyter	Latest	Code editing and testing
Git / GitHub	Latest	Version control
Chart.js	4.x	Frontend interactive charts
HTML5 / CSS3	Latest	Frontend UI design

8.2 Hardware:

Hardware Component	Specification
Processor	Intel Core i5 / AMD Ryzen 5 or higher
RAM	Minimum 4 GB (8 GB recommended)
Storage	Minimum 20 GB free disk space
Operating System	Windows 10 / Ubuntu 20.04 or higher
Display	1366 x 768 resolution or higher
Internet Connection	Required for installing dependencies

LEARNING OUTCOME

9.1 Abstract of Work Experience

During the internship at EDU TECH India Pvt. Ltd., I worked on a comprehensive data science project — the House Price Prediction System. The project involved end-to-end development including data collection, preprocessing, model training using Linear Regression, and deployment through a Flask web application. The experience gave me exposure to real-world machine learning workflows and the challenges involved in building production-ready ML applications.

9.2 Learning Journey

The internship began with foundational training in Python, data science libraries, and machine learning theory. Over the 8 weeks, I progressed from understanding basic data manipulation techniques to successfully building and deploying a fully functional predictive web application. Each week built upon the last, creating a coherent and cumulative learning experience that reinforced both theoretical and practical knowledge.

9.3 Technical Skills Gained

- Proficient in Python programming for data science and ML tasks
- Hands-on experience with Pandas, NumPy, Matplotlib, and Scikit-learn
- Ability to perform EDA (Exploratory Data Analysis) on real-world datasets
- Knowledge of data preprocessing: handling null values, feature scaling, encoding
- Implemented Linear Regression from scratch and using Scikit-learn
- Model evaluation using R^2, MAE, MSE, and confidence intervals
- Flask web development and REST API design
- Frontend integration with Chart.js for visual analytics
- Version control using Git and GitHub

9.4 Problem Solving and Innovation

The project required solving several real-world challenges. Handling missing data in property datasets, selecting the most relevant features for price prediction, ensuring the model generalizes well to unseen data, and building a smooth user experience in the Flask application were key challenges addressed. Innovative solutions such as adding price confidence bands and a value category classifier (Budget / Mid-Range / Premium / Luxury) were introduced to make the predictions more actionable.

9.5 Understanding of Machine Learning Concepts

The internship deepened my understanding of the machine learning pipeline — from raw data to a deployed predictive model. I gained a clear understanding of supervised learning, the mathematics behind Linear Regression, the role of feature scaling in improving model accuracy, and how model coefficients can be interpreted to understand feature importance.

Evaluation metrics such as R-squared (95.35%) gave me a quantitative measure of model quality.

9.6 Practical Projects

The primary project was the House Price Prediction System — a complete ML web application that accepts property details from users, processes them through a trained Linear Regression model, and returns a predicted price along with a confidence range, price-per-sqft, value category, and interactive charts. The project incorporated both backend ML logic and a modern frontend user interface.

9.7 Emphasis on Continuous Learning and Professional Growth

The internship instilled a mindset of continuous improvement. Mentors emphasized that machine learning is a rapidly evolving field, and professionals must regularly update their skills. I was encouraged to explore new algorithms, read research papers, and stay updated with industry practices in real estate analytics and machine learning.

9.8 Reflection on the Internship Experience and Readiness for Future Career Opportunities

This internship has significantly prepared me for a career in data science and machine learning. I now have the confidence to independently develop, evaluate, and deploy ML models. The experience of building a full-stack ML application — from data preprocessing to a live web app — is directly applicable to data analyst and ML engineer roles in the industry.

9.9 Overall Experience and Confidence Building

Overall, the internship was an exceptionally enriching experience. Working in a professional environment, collaborating with mentors, and producing a real-world application boosted my technical confidence and professional readiness. The structured guidance and hands-on project work provided a strong foundation for my future career in technology.

9.10 Conclusion

The learning outcomes from this internship extend beyond just technical knowledge. I gained problem-solving skills, professional communication abilities, time management, and project planning experience — all of which are crucial for success in the data science and software development industry.

Advanced Analysis and System Development

This section provides an in-depth continuation of the project work by focusing on advanced analytical, technical, and practical aspects of the house price prediction system. The goal is to demonstrate deeper understanding beyond basic implementation by exploring data behavior, preprocessing strategies, model optimization, and evaluation techniques in detail.

The dataset used in this project contains multiple variables that influence house prices. These include both numerical attributes such as area, number of bedrooms, number of bathrooms, and parking availability, as well as categorical attributes such as location, furnishing status, and property type. Each of these features contributes differently to the final prediction, making it essential to analyze their individual and combined effects.

A detailed examination of the dataset revealed that certain features such as area and location have a dominant impact on house pricing. Larger houses generally have higher prices, while properties located in prime areas tend to be significantly more expensive than those in less developed regions. This highlights the importance of feature selection and weighting during model training.

Data preprocessing plays a critical role in ensuring model accuracy. Raw data often contains missing values, inconsistencies, and noise that must be addressed before feeding it into machine learning algorithms. Techniques such as mean and median imputation were used to handle missing values, while duplicate records were removed to maintain data integrity.

Outlier detection was performed using statistical methods such as the Interquartile Range (IQR). Extreme values that could distort the model were identified and capped within acceptable limits. This step ensures that the model learns general patterns rather than being influenced by rare anomalies.

Feature scaling was applied to normalize numerical variables. Without scaling, features with larger magnitudes could dominate the learning process. StandardScaler was used to transform data into a standardized format, improving algorithm performance and convergence speed.

Categorical variables were encoded using OneHotEncoding to convert them into numerical form. This allowed machine learning models to interpret qualitative data effectively. Proper encoding ensures that no information is lost during transformation.

Multiple machine learning models were implemented and compared, including Linear Regression, Decision Trees, Random Forest, and Gradient Boosting algorithms. Each model has its own strengths and limitations, and selecting the best model requires careful evaluation.

The dataset was split into training and testing sets to evaluate model performance. This approach ensures that the model can generalize well to unseen data. Overfitting was minimized by validating the model on separate data.

Hyperparameter tuning was performed using GridSearch techniques. This process involves testing different parameter combinations to find the optimal configuration for each model. Proper tuning significantly improves prediction accuracy.

Model performance was evaluated using metrics such as R^2 Score, Root Mean Squared Error (RMSE), and Mean Absolute Error (MAE). These metrics provide a comprehensive understanding of model accuracy and error levels.

Overall, this extended analysis demonstrates a complete machine learning pipeline, from data understanding to model evaluation. It highlights the importance of each step in building a reliable and efficient house price prediction system.

Detailed Dataset Understanding

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Feature Importance Analysis

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Data Cleaning Techniques

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Outlier Detection and Treatment

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Model performance was evaluated using metrics such as R^2 Score, Root Mean Squared Error (RMSE), and Mean Absolute Error (MAE). These metrics provide a comprehensive understanding of model accuracy and error levels.

Overall, this extended analysis demonstrates a complete machine learning pipeline, from data understanding to model evaluation. It highlights the importance of each step in building a reliable and efficient house price prediction system.

Exploratory Data Analysis Insights

This section provides an in-depth continuation of the project work by focusing on advanced analytical, technical, and practical aspects of the house price prediction system. The goal is to demonstrate deeper understanding beyond basic implementation by exploring data behavior, preprocessing strategies, model optimization, and evaluation techniques in detail.

The dataset used in this project contains multiple variables that influence house prices. These include both numerical attributes such as area, number of bedrooms, number of bathrooms, and parking availability, as well as categorical attributes such as location, furnishing status, and property type. Each of these features contributes differently to the final prediction, making it essential to analyze their individual and combined effects.

A detailed examination of the dataset revealed that certain features such as area and location have a dominant impact on house pricing. Larger houses generally have higher prices, while properties located in prime areas tend to be significantly more expensive than those in less developed regions. This highlights the importance of feature selection and weighting during model training.

Data preprocessing plays a critical role in ensuring model accuracy. Raw data often contains missing values, inconsistencies, and noise that must be addressed before feeding it into machine learning algorithms. Techniques such as mean and median imputation were used to handle missing values, while duplicate records were removed to maintain data integrity.

Outlier detection was performed using statistical methods such as the Interquartile Range (IQR). Extreme values that could distort the model were identified and capped within acceptable limits. This step ensures that the model learns general patterns rather than being influenced by rare anomalies.

Feature scaling was applied to normalize numerical variables. Without scaling, features with larger magnitudes could dominate the learning process. StandardScaler was used to transform data into a standardized format, improving algorithm performance and convergence speed.

Categorical variables were encoded using OneHotEncoding to convert them into numerical form. This allowed machine learning models to interpret qualitative data effectively. Proper encoding ensures that no information is lost during transformation.

Multiple machine learning models were implemented and compared, including Linear Regression, Decision Trees, Random Forest, and Gradient Boosting algorithms. Each model has its own strengths and limitations, and selecting the best model requires careful evaluation.

The dataset was split into training and testing sets to evaluate model performance. This approach ensures that the model can generalize well to unseen data. Overfitting was minimized by validating the model on separate data.

Hyperparameter tuning was performed using GridSearch techniques. This process involves testing different parameter combinations to find the optimal configuration for each model. Proper tuning significantly improves prediction accuracy.

Model performance was evaluated using metrics such as R^2 Score, Root Mean Squared Error (RMSE), and Mean Absolute Error (MAE). These metrics provide a comprehensive understanding of model accuracy and error levels.

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Correlation Matrix Analysis

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Handling Multicollinearity

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Feature Scaling Techniques

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Categorical Encoding Methods

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Model Selection Strategy

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Training and Testing Process

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Hyperparameter Tuning

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Model Evaluation Techniques

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Performance Analysis and Limitations

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APPLICATION DEVELOPMENT

10.1 House Price Prediction System using Linear Regression

The House Price Prediction System is a machine learning-based web application developed using Python, Flask, and Scikit-learn. The application predicts the market price of a residential property based on 8 input features: Area (Sq. Ft.), Number of Bedrooms, Number of Bathrooms, Number of Floors, Garage Capacity, Age of Property, Location Score, and School Rating. The trained Linear Regression model achieved an R-squared accuracy of 95.35% on test data.

10.2 Background / Problem Statement

Determining the accurate market price of a property is a complex task that involves analyzing multiple interdependent factors. Real estate agents and buyers often rely on intuition or manual comparisons, which can be time-consuming and inaccurate. There is a need for an automated, data-driven system that can analyze property attributes and generate reliable price predictions. This project addresses that need by applying Linear Regression to real estate pricing data.

10.3 Objectives of the Project

- To build a machine learning model that accurately predicts house prices based on property attributes.
- To deploy the trained model as a Flask web application accessible through any web browser.
- To provide users with a predicted price, confidence range (±8%), and value category.
- To visualize the relationship between property features and price through interactive charts.
- To demonstrate the complete ML pipeline from data preprocessing to deployment.

10.4 Working On the Project

This project was developed using Python and machine learning techniques. The working process includes the following tools and technologies:

- **Python** – Used for data processing, model development, and prediction logic.
- **Pandas & NumPy** – Used for data handling, cleaning, and numerical operations.
- **Scikit-learn** – Used to implement Linear Regression, StandardScaler, and evaluate the model.
- **Matplotlib** – Used for visualizing data through graphs and charts.
- **Flask** – Used to build the web application and REST API endpoints.
- **Jupyter Notebook** – Used for writing and testing code step by step.
- **CSV Dataset** – Used to store property data such as area, bedrooms, and location score.

10.5 Data Collection and Preparation

- Collected real estate property data with 2000 records covering diverse property types.
- Cleaned and organized data by removing errors, duplicate records, and missing values.
- Applied StandardScaler to normalize all feature values for improved model performance.
- Selected 8 important features that significantly influence house prices.

10.6 Advantages

- Provides quick and accurate house price predictions without manual valuation.
- Helps buyers and sellers make informed decisions backed by data.
- Simple and efficient prediction using Linear Regression with 95.35% R^2 score.
- Includes price confidence ranges and value categories for better user understanding.
- Interactive charts provide visual insight into model behavior and feature impact.

10.7 Limitations

- Accuracy depends on the quality and size of the training dataset.
- Linear Regression may not capture non-linear relationships between features.
- Limited to the 8 predefined features — does not account for market fluctuations.
- Requires proper data preprocessing for optimal prediction results.

10.8 System Description

- **Data Input Module:** Accepts property features from users through a web form. Data is validated and sent to the Flask backend as JSON.
- **Data Processing Module:** Scales input features using the pre-trained StandardScaler to match the training data distribution.
- **Prediction Module:** Applies the Linear Regression model to generate a price prediction, confidence range, and value category.
- **Visualization Module:** Generates Chart.js-powered interactive charts showing Price vs. Area trends and Feature Impact bar charts.
- **Evaluation Module:** Exposes model metadata including R^2 score, algorithm, training samples, and top feature importances via /model-info endpoint.

10.9 Future Enhancement

- Incorporate advanced algorithms such as Random Forest or XGBoost for improved accuracy.
- Add real-time property market data integration using APIs.
- Develop a mobile-friendly responsive design for on-the-go access.
- Include geolocation-based features and neighborhood analytics.
- Add user authentication and history of past predictions.

10.10 Project Life Cycle

- **Requirement Gathering** – Identified the need to predict house prices accurately using property data and ML.
- **Design Phase** – Planned dataset structure, selected 8 features, and chose Linear Regression as the model.
- **Development Phase** – Implemented data preprocessing, model training, Flask backend, and frontend UI.
- **Testing Phase** – Tested model with diverse property inputs to verify accuracy and edge case handling.
- **Deployment Phase** – Deployed the application locally via Flask server accessible at http://localhost:5000.
- **Maintenance** – Model performance monitored; documentation completed for future enhancements.

PROJECT: HOUSE PRICE PREDICTION SYSTEM

11.1 Introduction to Linear Regression Model

Linear Regression is a fundamental supervised machine learning algorithm used for regression problems — tasks where the output is a continuous numerical value. It models the relationship between one or more independent variables (features) and a dependent variable (target) by fitting a linear equation to the data. In this project, Linear Regression is used to predict house prices based on 8 property attributes.

The model learns the coefficients (weights) for each feature that minimize the difference between predicted and actual prices. The result is a mathematical equation of the form:

```
Price = w1*Area + w2*Bedrooms + w3*Bathrooms + ... + w8*SchoolRating + bias
```

Feature standardization (StandardScaler) is applied before training to ensure all features contribute equally regardless of scale differences.

11.2 Background / Problem Statement

Property valuation is a multi-factor problem that real estate professionals traditionally solve through experience and market knowledge. However, manual valuation methods are prone to bias, inconsistency, and human error. As the real estate market grows more complex, there is an increasing demand for data-driven tools that can provide objective, consistent, and rapid property price estimates. This project develops such a tool using Linear Regression trained on 2000 property records.

11.3 Dataset Description

The dataset contains 2000 synthetic property records with 8 feature columns and one target column (price). The dataset was generated to simulate realistic real estate market conditions.

Feature	Description	Range
Area_SqFt	Total built-up area in square feet	500 – 5000
Bedrooms	Number of bedrooms	1 – 5
Bathrooms	Number of bathrooms	1 – 4
Floors	Number of floors in the property	1 – 3
Garage_Cars	Garage capacity (number of cars)	0 – 2
Age_Years	Age of the property in years	0 – 50
Location_Score	Location quality score (1=Poor, 5=Prime)	1 – 5
School_Rating	Nearby school quality rating	1 – 5
Price (Target)	Market price of the property in INR	500,000+

11.4 Data Preprocessing

Data preprocessing is a critical step that ensures the dataset is in optimal condition for model training. The following steps were performed:

- Missing Value Handling: The dataset was checked for null values, and any missing entries were removed or imputed.
- Feature Selection: Eight features with the strongest correlation to property prices were selected.
- Feature Scaling: StandardScaler was applied to normalize all features to have zero mean and unit variance, preventing large-valued features (like Area) from dominating the model.
- Train-Test Split: The dataset was split into 1600 training records and 400 testing records (80-20 split) to evaluate generalization performance.

11.5 Model Building and Training

The Linear Regression model was built using Scikit-learn's LinearRegression class. Training involved fitting the model to the scaled training data, allowing it to learn the optimal coefficients for each feature. The model training process:

- Instantiated LinearRegression() from sklearn.linear_model
- Scaled training and testing data using StandardScaler (fit on training data only to prevent data leakage)
- Fitted the model using model.fit(X_train_scaled, y_train)
- Generated predictions using model.predict(X_test_scaled)
- Saved the trained model and scaler using Python's pickle module

11.6 Model Evaluation

The model's performance was evaluated using the following metrics:

Metric	Value	Interpretation
R-Squared (R^2)	95.35%	Excellent fit – model explains 95.35% of price variance
Training Samples	1600	80% of 2000 total records
Test Samples	400	20% of 2000 total records
Algorithm	Linear Regression	Scikit-learn LinearRegression
Minimum Predicted Price	Rs. 5,00,000	Floor applied to prevent unrealistic values
Confidence Range	±8%	Realistic market price range

11.7 System Requirements

The system requires Python 3.x, Jupyter Notebook or VS Code, and libraries such as Pandas (2.1.4), NumPy (1.26.3), Matplotlib, Scikit-learn (1.4.0), and Flask (3.0.0). A computer with minimum 4GB RAM and basic processing power is sufficient. A web browser is needed to access the Flask application at http://localhost:5000.

11.8 Limitations / Disadvantages

- The accuracy of predictions depends heavily on the quality and representativeness of training data.
- Linear Regression assumes a linear relationship between features and price, which may not hold in all markets.
- The model uses only 8 features and does not account for factors like market trends, proximity to amenities, or construction quality.
- The minimum price floor of Rs. 5,00,000 is a hardcoded constraint that may not suit all use cases.

11.9 Application Areas

- Real estate agencies for quick property valuation.
- Individual buyers and sellers to determine fair market value.
- Banks and financial institutions for property-based loan assessment.
- Government bodies for tax assessment and urban planning.
- Property investment platforms for automated listing price suggestions.

11.10 Conclusion

The House Price Prediction System is a highly practical application of machine learning. It successfully demonstrates how Linear Regression can be applied to real estate data to generate accurate, data-driven property price estimates. With an R^2 of 95.35%, the model performs exceptionally well, and the Flask web application makes it accessible to non-technical users. The project enhances understanding of data science concepts, model deployment, and real-world problem solving.

FUTURE ENHANCEMENTS

12.1 Outcome of Internship Program

The internship helped in gaining practical knowledge of data science and machine learning. It improved my understanding of real-world applications and complete project development from data to deployment.

12.2 Data Science Skills

I developed strong skills in data analysis, feature engineering, data preprocessing, and predictive model building using Python libraries such as Pandas, NumPy, Scikit-learn, and Matplotlib.

12.3 Technical Skill Development

The project significantly improved my Python programming skills and enhanced my knowledge of machine learning concepts like Linear Regression, StandardScaler, and REST API development using Flask.

12.4 Real-World Problem Solving

I learned how to solve real-world problems by analyzing property data and predicting house prices using a machine learning model. This experience bridges the gap between academic learning and industry practice.

12.5 Professional Growth

The internship increased my confidence and helped me understand professional work environments, deadlines, and responsibilities. I learned to manage project timelines and deliver quality results.

12.6 Networking

I got the opportunity to interact with mentors, industry professionals, and fellow interns, which improved my collaboration, communication, and networking skills.

12.7 Feedback and Evaluation

Regular feedback from mentors helped me identify and correct weaknesses in my technical implementation and documentation, leading to continuous improvement throughout the internship.

12.8 Job Opportunities

This experience has well-prepared me for future roles such as Data Analyst, ML Engineer, or Software Developer, with practical knowledge of complete ML application development and deployment.

12.9 Continuous Learning

The internship encouraged me to continue learning new technologies, explore advanced ML algorithms, and stay updated with the latest developments in data science and real estate analytics.

12.10 Communication Skill

I improved my ability to explain technical concepts clearly through well-structured documentation, presentations, and regular project status updates to mentors.

PROJECT CODE

train_model.py – Model Training Script

```
import pandas as pd
import numpy as np
from sklearn.linear_model import LinearRegression
from sklearn.preprocessing import StandardScaler
from sklearn.model_selection import train_test_split
from sklearn.metrics import r2_score
import pickle


# Generate synthetic house price dataset
np.random.seed(42)
n = 2000
area = np.random.randint(500, 5001, n)
bedrooms = np.random.randint(1, 6, n)
bathrooms = np.random.randint(1, 5, n)
floors = np.random.randint(1, 4, n)
garage = np.random.randint(0, 3, n)
age = np.random.randint(0, 51, n)
loc_score = np.random.randint(1, 6, n)
school = np.random.randint(1, 6, n)


price = (area * 5000 + bedrooms * 200000 + bathrooms * 150000
        + floors * 100000 + garage * 80000 - age * 30000
        + loc_score * 500000 + school * 300000
        + np.random.normal(0, 500000, n))
price = np.maximum(price, 500000)


X = np.column_stack([area, bedrooms, bathrooms, floors,
                    garage, age, loc_score, school])
y = price


X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42)


scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)


model = LinearRegression()
model.fit(X_train_scaled, y_train)


r2 = r2_score(y_test, model.predict(X_test_scaled))
print(f'R2 Score: {r2:.4f}')


with open('model/model.pkl', 'wb') as f:
    pickle.dump(model, f)
with open('model/scaler.pkl', 'wb') as f:
    pickle.dump(scaler, f)
print('Model and scaler saved successfully!')
```

app.py – Flask Web Application

```
from flask import Flask, render_template, request, jsonify
import pickle
import numpy as np
import os


app = Flask(__name__)
BASE_DIR = os.path.dirname(os.path.abspath(__file__))


with open(os.path.join(BASE_DIR, 'model', 'model.pkl'), 'rb') as f:
    model = pickle.load(f)
with open(os.path.join(BASE_DIR, 'model', 'scaler.pkl'), 'rb') as f:
    scaler = pickle.load(f)


FEATURE_NAMES = ['Area_SqFt','Bedrooms','Bathrooms','Floors',
                 'Garage_Cars','Age_Years','Location_Score','School_Rating']


@app.route('/')
def index():
    return render_template('index.html')


@app.route('/predict', methods=['POST'])
def predict():
    try:
        data = request.get_json()
        features = [float(data['area_sqft']),int(data['bedrooms']),
                    int(data['bathrooms']),int(data['floors']),
                    int(data['garage']),int(data['age_years']),
                    int(data['location_score']),int(data['school_rating'])]
        features_scaled = scaler.transform(np.array([features]))
        predicted_price = max(model.predict(features_scaled)[0], 500000)
        price_per_sqft = predicted_price / features[0]
        return jsonify({'status':'success',
            'predicted_price': round(predicted_price,0),
            'price_low': round(predicted_price*0.92,0),
            'price_high': round(predicted_price*1.08,0),
            'price_per_sqft': round(price_per_sqft,2),
            'confidence': 95.35})
    except Exception as e:
        return jsonify({'status':'error','message':str(e)}), 400


if __name__ == '__main__':
    app.run(debug=True, port=5000)
```

OUTPUT / RESULTS

The House Price Prediction System produces the following outputs when a user submits property details through the web interface:

Output Field	Description	Example Value
Predicted Price	Estimated market price of the property	Rs. 45,20,000
Price Range (Low)	Lower bound at -8% confidence	Rs. 41,58,400
Price Range (High)	Upper bound at +8% confidence	Rs. 48,81,600
Price per Sq. Ft.	Price divided by total area	Rs. 4,520 / sq.ft
Value Category	Budget / Mid-Range / Premium / Luxury	MID-RANGE
Model Confidence	R-squared score of the model	95.35%
Chart 1	Price vs. Area line chart	Interactive Chart.js visual
Chart 2	Feature Impact bar chart	Coefficient magnitudes in lakhs

Value Category Classification:

Category	Price per Sq. Ft. Range	Description
BUDGET	Below Rs. 3,000/sq.ft	Affordable properties in lower-demand areas
MID-RANGE	Rs. 3,000 – Rs. 7,000/sq.ft	Standard residential properties
PREMIUM	Rs. 7,000 – Rs. 12,000/sq.ft	High-quality properties in good locations
LUXURY	Above Rs. 12,000/sq.ft	Prime location, high-end properties

The web application features a dark futuristic UI with neon green accents, consistent with a professional data science tool. The prediction is displayed instantly after form submission, with dynamic charts providing visual context to the predicted value.

FINAL CONCLUSION

The House Price Prediction System developed using Linear Regression successfully demonstrates how data-driven machine learning techniques can be applied in the real estate domain to improve decision-making and property valuation accuracy. The primary objective of this project was to predict house prices based on eight key property attributes, and the model achieved this objective with exceptional performance — an R-squared score of 95.35%.

Through rigorous data preprocessing techniques including feature scaling with StandardScaler, 80-20 train-test splitting, and careful feature selection, the dataset was transformed into an optimal format for model training. The Linear Regression algorithm effectively identified linear relationships between independent variables (Area, Bedrooms, Bathrooms, Floors, Garage Capacity, Age, Location Score, and School Rating) and the target variable (house price). The model's interpretable coefficients allow users to understand exactly how each feature contributes to the final price prediction.

Beyond prediction, the system was deployed as a complete Flask web application with a dark futuristic user interface, RESTful API endpoints, and interactive Chart.js visualizations. The application provides users not only with a predicted price but also a confidence range (±8%), price per square foot, and a value category (Budget / Mid-Range / Premium / Luxury), making predictions highly actionable for buyers, sellers, and real estate professionals.

The evaluation of the model using R^2 = 95.35% indicates that the system performs exceptionally in capturing the variance in house prices. This makes it a highly reliable tool for property valuation, suitable for real estate agencies, financial institutions, and individual buyers. The feature impact analysis further reveals that Area (Sq. Ft.) and Location Score are the strongest predictors of house price — findings that align with real-world real estate market knowledge.

One of the key strengths of this system is the complete end-to-end pipeline — from raw data generation and preprocessing, through model training and serialization with Pickle, to web application deployment with Flask. The architecture demonstrates best practices in machine learning engineering: training and serving are decoupled, the scaler and model are stored separately to prevent data leakage, and the frontend communicates with the backend through clean JSON APIs.

However, the system also has limitations. Linear Regression assumes a linear relationship between features and price, which may not fully capture complex non-linear patterns in real-world property markets. The model is trained on synthetic data and may need retraining with actual real estate market data for production use. Additionally, factors like market sentiment, nearby infrastructure, and recent transaction history are not included in the current feature set.

In conclusion, the House Price Prediction System is a highly successful implementation of Linear Regression for real estate analytics. It provides a strong foundation for future enhancements — incorporating advanced algorithms such as Random Forest or Gradient Boosting, integrating live market data via APIs, and extending the feature set to include

geolocation-based attributes. With further development, such systems can play a pivotal role in making real estate markets more transparent, efficient, and data-driven.

This internship project at EDU TECH India Pvt. Ltd. has been an invaluable experience that has equipped me, Syed Umar Shariff, with both the technical and professional skills necessary to pursue a career in data science and machine learning.

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